

C657

Calculus III

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A tutor who has taught calculus for 15 years assigns the series $\sum n^2/3^n$, starting at $n = 1$. The first three terms are $1/3$, $4/9$, and $1/3$. What does the ratio test show?

- A. $L = 1$, so the test is inconclusive.
- B. The series converges to $1/3$.
- C. $L = 1/3$, so the series converges.
- D. The series diverges because the second term is larger than the first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Applying the ratio test to $\sum 1/n^2$ gives $L = 1$. What does the ratio test itself allow you to conclude?

- A. The series converges because its terms go to zero.
- B. The series diverges because L is not less than 1.
- C. The series converges because $L \leq 1$.
- D. Nothing; another test is needed (here, the p-series test shows convergence).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

On a practice set written by a department of six teachers, a student must pick the series that converges. The set is due Friday. Which one converges?

- A. $\sum 1/\sqrt{n}$, because $p = 1/2$ is positive
- B. $\sum n/(n + 1)$
- C. $\sum 1/n$, because its terms approach zero
- D. $\sum 1/n^{1.1}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An engineer uses a Maclaurin polynomial of $e^{(2x)}$ to estimate a quantity in a battery model developed over two years. The model runs on a laptop with 16 GB of memory. What is the coefficient of x^3 ?

- A. $4/3$
- B. 8
- C. $1/6$
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a robotics club that meets twice a week, a student models two support rods as $u = \langle 1, 2, 3 \rangle$ and $v = \langle 4, -5, 2 \rangle$. The rods are each 40 cm long in the build. What is the angle between the vectors?

- A. 0° , because the dot product is zero
- B. 90°
- C. Greater than 90° , because v has a negative component
- D. It cannot be found without the cross product.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

For what value of k are $\langle 2, k, 1 \rangle$ and $\langle 3, -2, 4 \rangle$ perpendicular?

- A. 10
- B. -5
- C. 3
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A designer at a firm with offices in three cities lays out a parallelogram-shaped panel with edge vectors $u = \langle 2, 1, 0 \rangle$ and $v = \langle 1, 4, 0 \rangle$ in meters. The panel will be painted blue. What is its area?

- A. 49 square meters
- B. 7 square meters
- C. 3.5 square meters
- D. 6 square meters

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8.

An engineer modeling a solar panel mount installed in 2021 needs the plane through the point $(1, 0, 2)$ with normal vector $\langle 2, -1, 3 \rangle$. The mount faces south. Which equation describes the plane?

- A. $x + 2z = 8$
- B. $2x - y + 3z = 8$
- C. $\langle 1, 0, 2 \rangle + t\langle 2, -1, 3 \rangle$
- D. $2x - y + 3z = 0$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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9.

A hiker's elevation is modeled by $f(x, y) = x^2y$ on a trail map printed in 2019. At the point $(1, 2)$, what is the rate of change of f toward the direction of the vector $\langle 3, 4 \rangle$? The hiker has already walked 6 km.

- A. $\sqrt{17} \approx 4.12$
- B. 16
- C. 3.2
- D. $\langle 4, 1 \rangle$

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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10. [TRANSFER — reversed stem / different variable]

For $f(x, y) = x^2y$ at $(1, 2)$, in what direction does f increase fastest, and at what rate?

- A. Direction $\langle 4, 1 \rangle$, rate $\sqrt{17}$
- B. Direction $\langle 4, 1 \rangle$, rate 17
- C. Direction $\langle 1, 2 \rangle$, rate $\sqrt{5}$
- D. Direction $\langle -4, -1 \rangle$, rate $\sqrt{17}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A cost model built by a team of four analysts is $f(x, y) = x^2 + 4xy + y^2$, with a critical point at $(0, 0)$. The team presented it last Tuesday. How should $(0, 0)$ be classified?

- A. Local minimum, because f_{xx} and f_{yy} are positive
- B. Local maximum
- C. Saddle point
- D. Inconclusive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

At a critical point, $f_{xx} = 2$ and $f_{yy} = 8$. For which values of f_{xy} is the point a local minimum?

- A. $|f_{xy}| < 4$
- B. $|f_{xy}| > 4$
- C. $|f_{xy}| \leq 4$
- D. $f_{xy} < 16$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A tutor ~~who has taught calculus for 15 years~~ assigns the series $\sum n^2/3^n$, starting at $n = 1$. ~~The first three terms are 1/3, 4/9, and 1/3.~~ What does the ratio test show?

Target word: convergence by ratio test **Decides:** The ratio $(n+1)^2/(3n^2)$ approaches 1/3.

A	Dropped element	Drops the 1/3 from the exponential part of the ratio.
B	Adjacent category	L is the limit of the ratio, not the sum of the series.
C	Correct	The polynomial ratio tends to 1 and the 1/3 remains; $L < 1$.
D	Camouflage	Early terms do not decide convergence; the limit of the ratio does.

2. Answer D Concept 1 · transfer

Applying the ratio test to $\sum 1/n^2$ gives $L = 1$. What does the ratio test itself allow you to conclude?

Target word: meaning of $L = 1$ **Decides:** $L = 1$ is the inconclusive case.

A	Confusion set	Terms going to zero never proves convergence.
B	Inverted operation	Divergence requires $L > 1$.
C	Absolute language	Only L strictly less than 1 guarantees convergence.
D	Correct	The ratio test is silent at $L = 1$.

3. Answer D Concept 2

On a practice set ~~written by a department of six teachers~~, a student must pick the series that converges. ~~The set is due Friday.~~ Which one converges?

Target word: pick the convergent series **Decides:** p-series converge only for $p > 1$.

A	Inverted operation	p must exceed 1; $p = 1/2$ diverges.
B	Dropped element	Its terms approach 1, so it fails the nth-term test and diverges.
C	Confusion set	The harmonic series diverges even though its terms go to zero.
D	Correct	$p = 1.1 > 1$.

4. Answer A Concept 3

An engineer uses a Maclaurin polynomial of $e^{(2x)}$ to estimate a quantity in a battery model ~~developed over two years. The model runs on a laptop with 16 GB of memory.~~ What is the coefficient of x^3 ?

Target word: one coefficient **Decides:** The x^3 term is $(2x)^3/3! = 8x^3/6$.

A	Correct	$8/6 = 4/3$.
B	Dropped element	Leaves out the 3! in the denominator.
C	Confusion set	That is the coefficient for e^x , ignoring the factor 2.
D	Wrong stem number	Divides 8 by 2! instead of 3!.

5. Answer B Concept 4

In a robotics club ~~that meets twice a week~~, a student models two support rods as $u = \langle 1, 2, 3 \rangle$ and $v = \langle 4, -5, 2 \rangle$. ~~The rods are each 40 cm long in the build.~~ What is the angle between the vectors?

Target word: angle between vectors **Decides:** $u \cdot v = 4 - 10 + 6 = 0$.

A	Inverted operation	Parallel vectors have $ u \cdot v = u v $, not zero.
B	Correct	A zero dot product means perpendicular.
C	Confusion set	One negative component does not fix the sign of the dot product.
D	Adjacent category	The dot product alone settles the angle here.

6. Answer D Concept 4 • transfer

For what value of k are $\langle 2, k, 1 \rangle$ and $\langle 3, -2, 4 \rangle$ perpendicular?

Target word: solve for a component **Decides:** $6 - 2k + 4 = 0$.

A	Wrong stem number	Stops at the constant sum without dividing by 2.
B	Inverted operation	Sign error when solving.
C	Dropped element	Leaves out the z -components (1×4).
D	Correct	$10 - 2k = 0$ gives $k = 5$.

7. Answer B Concept 5

A designer ~~at a firm with offices in three cities~~ lays out a parallelogram-shaped panel with edge vectors $u = \langle 2, 1, 0 \rangle$ and $v = \langle 1, 4, 0 \rangle$ in meters. ~~The panel will be painted blue.~~ What is its area?

Target word: parallelogram area **Decides:** $u \times v = \langle 0, 0, 7 \rangle$.

A	Dropped element	Squared magnitude without the square root.
B	Correct	$ u \times v = 7$.
C	Adjacent category	That is the triangle area, half the parallelogram.
D	Confusion set	That is the dot product, $2 + 4$.

8. Answer B Concept 6

An engineer modeling a solar panel mount ~~installed in 2021~~ needs the plane through the point $(1, 0, 2)$ with normal vector $\langle 2, -1, 3 \rangle$. ~~The mount faces south.~~ Which equation describes the plane?

Target word: plane equation **Decides:** $2(x - 1) - (y - 0) + 3(z - 2) = 0$.

A	Inverted operation	Uses the point as the normal and the normal as the point.
B	Correct	Expanding gives $2x - y + 3z - 8 = 0$.
C	Adjacent category	That is the normal line, not the plane.
D	Dropped element	Ignores the point, so the plane passes through the origin.

9. Answer C Concept 7

A hiker's elevation is modeled by $f(x, y) = x^2y$ on a trail map printed in 2019. At the point $(1, 2)$, what is the rate of change of f toward the direction of the vector $\langle 3, 4 \rangle$? ~~The hiker has already walked 6 km.~~

Target word: directional derivative **Decides:** $\nabla f(1, 2) = \langle 4, 1 \rangle$ and $u = \langle 0.6, 0.8 \rangle$.

A	Adjacent category	That is the maximum rate, in the gradient direction.
B	Dropped element	Dots with $\langle 3, 4 \rangle$ without making it a unit vector.
C	Correct	$4(0.6) + 1(0.8) = 3.2$.
D	Confusion set	That is the gradient vector, not a rate in a given direction.

10. Answer A Concept 7 • transfer

For $f(x, y) = x^2y$ at $(1, 2)$, in what direction does f increase fastest, and at what rate?

Target word: direction and rate of steepest ascent **Decides:** $\nabla f(1, 2) = \langle 4, 1 \rangle$.

A	Correct	Steepest ascent is along ∇f at rate $ \nabla f $.
B	Dropped element	Missing the square root in the magnitude.
C	Wrong stem number	Uses the point instead of the gradient.
D	Inverted operation	That is the direction of fastest decrease.

11. Answer C Concept 8

A cost model ~~built by a team of four analysts~~ is $f(x, y) = x^2 + 4xy + y^2$, with a critical point at $(0, 0)$. ~~The team presented it last Tuesday.~~ How should $(0, 0)$ be classified?

Target word: classify the critical point **Decides:** $D = (2)(2) - 4^2 = -12$.

A	Dropped element	Ignores the f_{xy} term in D .
B	Inverted operation	A maximum needs $D > 0$ and $f_{xx} < 0$.
C	Correct	$D < 0$.
D	Confusion set	Inconclusive only when $D = 0$.

12. Answer A Concept 8 • transfer

At a critical point, $f_{xx} = 2$ and $f_{yy} = 8$. For which values of f_{xy} is the point a local minimum?

Target word: solve for the mixed partial **Decides:** Need $D = 16 - (f_{xy})^2 > 0$ with $f_{xx} > 0$.

A	Correct	$16 - (f_{xy})^2 > 0$ means $ f_{xy} < 4$.
B	Inverted operation	That makes D negative: a saddle.
C	Absolute language	At $ f_{xy} = 4$, $D = 0$ and the test is inconclusive.
D	Dropped element	Forgets that f_{xy} is squared in D .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Ratio test	If $\lim a(n+1)/a(n) = L$, the series converges absolutely when $L < 1$, diverges when $L > 1$, and the test says nothing when $L = 1$.	Reading $L = 1$ as convergence, or treating L as the sum.	Factorials or powers like 3^n in the terms.
2	p-series and the nth-term test	$\sum 1/n^p$ converges only when $p > 1$; terms going to zero is necessary but not enough for convergence.	Concluding a series converges because its terms approach zero.	A power of n in the denominator, or the harmonic series.
3	Taylor and Maclaurin coefficients	The coefficient of $(x - a)^n$ is $f^{(n)}(a) / n!$.	Forgetting the $n!$ or the chain-rule factor inside the function.	A request for one specific coefficient of a series.
4	Dot product and orthogonality	$u \cdot v = u v \cos \theta$, so a dot product of zero means the vectors are perpendicular.	Confusing a zero dot product with parallel vectors.	Two vectors and a question about angle or perpendicularity.
5	Cross product magnitude	$ u \times v $ is the area of the parallelogram spanned by u and v ; the vector $u \times v$ is perpendicular to both.	Using the dot product for area, or halving to a triangle when a parallelogram is asked.	"Area" with two vectors in three dimensions.
6	Equation of a plane	A plane with normal (a, b, c) through (x_0, y_0, z_0) is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$.	Mixing up the point and the normal, or writing the normal line instead.	A normal vector and a point.
7	Gradient and directional derivative	$D_u f = \nabla f \cdot u$ with u a unit vector; the maximum rate of increase is $ \nabla f $ in the direction of ∇f .	Dotting with a non-unit direction vector.	"Rate of change in the direction of" or "steepest."
8	Second derivative test	$D = f_{xx}f_{yy} - (f_{xy})^2$: $D > 0$ with $f_{xx} > 0$ is a minimum, $D > 0$ with $f_{xx} < 0$ is a maximum, $D < 0$ is a saddle, $D = 0$ is inconclusive.	Calling a point a minimum because f_{xx} and f_{yy} are both positive while ignoring f_{xy} .	A critical point and all three second partials.

Course scope (public description): Calculus III is the study of calculus conducted in three-or-higher-dimensional space. It covers the knowledge and skills necessary to apply calculus of multiple variables while using the appropriate technology to model and solve real-life problems. Topics include: infinite series and convergence tests (integral, comparison, ratio, root, and alternating), power series, Taylor polynomials, vectors, lines and planes in three dimensions, dot and cross products, multivariable functions, limits, and continuity, partial derivatives, directional derivatives, gradients, tangent planes, normal lines, and extreme values. Calculus II is a prerequisite for this course.